

Utilising Machine Learning and Large Language Models for Mental Health Analysis on Social Media: A Dual-Context Approach for Evidence Summarisation and Risk Mitigation

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Utilising Machine Learning techniques with Large Language Models to produce a summary of a user's posts on social media and an appropriate response

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Abstract

This paper explores the combination of Machine Learning techniques and the use of Large Language Models (LLMs) in analysing the suicidal ideation of social media users. It focuses explicitly on highlighting evidence, producing summaries of posts and providing the user with a suitable response to mitigate the risk of suicide or self-harm. A dual-context approach utilises different contexts and methods to provide two LLMs, Tulu-2-DPO-7b and Mistral-7b-instruct-v0.3, the information required to produce summaries and responses. Each LLM's ability in this task, leveraging both chain-of-thought and zero-shot prompting methods, is compared and evaluated. Highlighting the use of meta-information and the dual-context approach demonstrates the ability of solutions in this context. The solution produces notable results in the evidence highlighting and summarisation tasks, and the new research in response generation is promising for future developments. A new precedent is set for research in mental health monitoring and advancing computational psychiatric care, aiming to reduce the number of annual deaths due to suicide.

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Chapter 1

Introduction

1.1 Background

Artificial Intelligence (AI) use in the healthcare sector is growing rapidly, receiving significant attention from researchers and medical professionals (Secinaro et al., 2021). The applications range from predictive models of medicine and the spread of diseases to diagnosing disease or injury and, more recently, in clinical decision-making. With such a wide range of applications, the specific needs vary substantially, and very different approaches are required.

Natural Language Processing (NLP) is a rapidly growing area of AI. It allows computers to better process and understand human language and perform significantly better in cognitive and creative tasks where it was previously inferior (Gruetzemacher, 2022). NLP is used in various healthcare tasks, particularly in tasks involving improving interactions and communications between patients and professionals (Locke et al., 2021).

A significant amount of unstructured patient data exists today, rapidly growing with the increase of reports and notes in the patient's medical history. This information can be complex for electronic systems to process and needs to be more organised for a medical professional to look at and understand. NLP techniques have the potential to help navigate these problems and improve patient care, as demonstrated by Iroju and Olaleke (2015) and their review.

NHS England recently rolled out a risk assessment tool called "Brave AI", which uses NLP techniques to identify individuals at risk of being hospitalised within the next year from the patient's medical records (NHS Somerset, 2023). Brave AI reduced care home residents' falls by 35%, their attendance to emergency departments by 60%, and ambulance callouts by 8.7%. These reductions demonstrate the ability of NLP to help medical professionals in risk prediction tasks involving large amounts of data.

1.1.1 Project Motivation

The World Health Organization reported that approximately 700,000 people die from suicide each year, equating to one person every 40 seconds, and there are indications that this is only a fraction of those attempting suicide (World Health Organization, 2021). NLP is used in everyday applications to try to prevent suicide attempts or get the proper support to those who need it. For example, companies like Google and Facebook have implemented algorithms that detect at-risk individuals from search queries for years, usually providing a local helpline

to talk to somebody. However, these are limited in use due to a range of modern language masking what might be considered as intent to self-harm or attempt suicide. Many of these algorithms are outdated and are in desperate need of a review and update.

Since the introduction of social media, there has been an ongoing rise in people using online forums, blogs and posts to communicate their mental health issues and suicidal tendencies (Park, Cha and Cha, 2012). This social media use poses an issue as these posts are often anonymous or through online identities, where friends and family cannot view them and offer the support the individual might need. Also, the anonymous nature of these websites has been shown to influence users into sharing their feelings and struggles (Lu, Pan and Deng, 2019), presenting an opportunity for online support.

With the rise of NLP and Machine Learning techniques, there has been extensive research into the detection of posts online where somebody might be suffering from mental health issues or contemplating self-harm or suicide. This project aims to investigate different Large Language Models, incorporating various Machine Learning methodologies to correctly detect social media users at risk of self-harm or suicide, providing an adequate and evidence-based summary of why the user is at risk and providing a suitable response.

1.2 Project Purpose

1.2.1 Project Aims

Within this project, the aims are:

- Compare different Machine Learning methods and the use of Large Language Models (LLMs) in detecting social media posts where a user is at risk of self-harm or suicide.
- Utilise the chosen models to assess social media posts, highlighting evidence and summarising the reasoning behind a generated risk level.
- Compare different setups and their summarisation ability.
- Develop and compare architectures allowing for an appropriate response to the user to mitigate the risk of suicide.

1.2.2 Project Objectives and Deliverables

- Conduct a literature review on the existing implementation of NLP and Machine Learning models in suicide risk detection in social media posts.
- Find and implement a social media post dataset, using pre-processing and data cleaning techniques to ensure the data is relevant and valuable.
- Combine several Machine Learning techniques to generate highlighted evidence of suicide risk in each user's posts.
- Test the evidence extraction model using relevant evaluation metrics.
- Utilise LLMs and prompting techniques to summarise each user's posts, incorporating their rating, evidence and explanation behind the rating allocation.

- Test the summarisation model using relevant evaluation metrics to compare different setups.
- Utilise LLMs and prompting methods to produce a response to the user based solely on the produced summary to offer support and mitigate risk.
- Test the response generation model using evaluation metrics, ensuring the response is helpful and empathetic.
- Complete a detailed analysis of the models and identify strengths and weaknesses, suggesting future improvements.

1.3 Structure of Dissertation

With well-defined aims, objectives and deliverables for the project, the structure of this paper will be as follows:

Literature and Technology Survey: Existing research on the implementation of different models in suicide risk detection will be examined and compared. The findings will be used to design the solution by evaluating the architectures and methods of solving similar tasks.

Methods: The overall task and solution will be discussed, the requirements defined, and the methods used to meet them explained.

Implementation: Each part of the solution's setup will be explained in more detail, including the testing completed at each stage to ensure that the outputs remain helpful and relevant.

Results: The overall results generated will be compared and scrutinised, prompting a discussion on why these results are significant.

Conclusion: The overall solution will be discussed and compared to the original aims and objectives, the project's limitations will be explained, and future work will be suggested.

1.4 Ethical Considerations

It is crucial to evaluate the task's nature at this point and consider its ethical requirements to uphold solid ethical responsibility in the field.

- The datasets used must not include information about persons that could allow for the identification of a user or mentioned person.
- No participants are used in the task, and any response generation will not be tested directly on patients.
- Transparency of all generated results is crucial to ensuring that the setup can be trusted and produce safe results.
- The dataset or sub-datasets will be appropriately cited, ensuring that the information is not used maliciously.

- The task is designed to improve specialist-patient interactions and should only be used with human oversight.

Chapter 2

Literature and Technology Survey

This literature review aims to find and compare existing Machine Learning techniques used in suicide risk detection in the text, specifically those used in analysing modern social media posts. The first section covers the development of Natural Language Processing architectures and their impact, highlighting the transition from Deep Learning to modern LLMs. Various similar tasks are then looked at, namely the CLPsych Shared Tasks, with different approaches compared and scrutinised to find the best methods for this task. The final section covers psychiatry and clinical mitigation to help guide the model in response generation.

2.1 NLP in determining suicide risk

2.1.1 Deep Learning

As early as 2018, Deep Learning was used to improve language modelling and text clarification in determining a suicide risk from social media (Coppersmith et al., 2018). This research involved a simple binary classification task, deciding whether a social media user was at risk of suicide through the examination of their posts. Unlike classic Deep Learning tasks, limited data was available, requiring both supervised and unsupervised learning methods to prevent the model from overfitting to training data.

The model was initialised using GloVe embeddings, a new log-bilinear regression model for unsupervised learning of word representations that vastly outperformed other models, reducing overfitting on the training set (Pennington, Socher and Manning, 2014). Sequences of word vectors were then processed via a bidirectional Long Short-Term Memory (LSTM) layer to capture the contextual information between each word. The output of each layer was combined into a single vector using skip connections into a self-attention layer, applying weights to the timesteps of the sentence. This layer meant that more informative sub-sequences were strongly considered. A softmax output linear layer led to a probability relating to the likelihood that the author of a given text was at risk of suicide.

Using these embeddings and new model layers led to a performance that was state of the art at the time for risk prediction from social media alone, showing the importance of these techniques in word representation. However, the model showed rapidly increasing false alarm rates as it was made more sensitive. For example, models with a 70-85% positive rate resulted in a 10% false alarm rate. The main issue revolved around the need for more context of the

wordings used within sentences, resulting in not being able to distinguish a sentence with a high risk of suicide from a sentence simply discussing the topic. This lack of context also meant it could not separately classify comments at a slight risk versus those at an imminent risk.

2.1.2 BERT

Later in 2018, there was a significant breakthrough in NLP thanks to the introduction of a new model called BERT (Devlin et al., 2018). In addition to the GloVe embedding technique used previously (Ziriky et al., 2019), BERT utilised the new transformer architecture (Vaswani, 2017). This innovation allowed the model to process all words in a given passage of text at once, using its self-attention mechanism to compute attention scores with every other token and itself. It could therefore simultaneously learn word context from both directions, dynamically assigning different weights to different sentence parts. BERT was a crucial development because it allowed for a better understanding of sentence meaning, enabling the model to distinguish between a user discussing suicide and a user at risk of suicide themselves.

The development of BERT models led to the development of Large Language Models (LLMs), which are used widely today as cutting-edge AI systems processing text coherently and generalising to multiple tasks (Radford et al., 2019). These models are a direct evolution of the previously used neural language modelling, trained in a self-supervised manner on a substantial volume of text. The aim of this was generic learning that could be used on a wide range of applications and NLP tasks and, with fine-tuning, could outperform previous language models (Naveed et al., 2023).

LLaMA is an LLM introduced by Meta in 2023, a collection of language models with billions of parameters released openly and outperforming other models while many times smaller (Touvron et al., 2023). These models were trained exclusively on publicly available data, and it was observed that fine-tuning these models on instructions led to promising results, as previously shown by Chung et al. (2024). Based on this model collection, MentaLLaMA was introduced using the Interpretable Mental Health Instruction (IMHI) dataset containing 105K data samples for enhanced instruction tuning and evaluation (Yang et al., 2023; Ouyang et al., 2022). Compared to other state-of-the-art methods and models, such as OpenAI's GPT-3, MentaLLaMA showed strong generalisability to unseen tasks and high correctness in predictions. These instruction-tuning methods outperformed other models across almost every test within the IMHI test set.

2.2 CLPsych

CLPsych is a workshop on Computational Linguistics and Clinical Psychology held annually as part of the Conference of the European Chapter of the Association for Computational Linguistics (CLPsych, 2024). As part of this workshop, a "Shared Task" is completed each year, focusing on improving computational approaches to psychology. Two of these tasks are presented in the following subsections, including their setup and how the performance is analysed. These are chosen as they are of a similar nature to the task being proposed in this paper.

2.2.1 Shared Task 2019

BERT models were first used in the domain of suicide risk detection when implemented as part of many approaches to the CLPsych-2019 shared task (Zirikly et al., 2019). This task involved a dataset collected from Reddit of all posts between 2005 and 2015, including 11,129 candidate users on the SuicideWatch subreddit, accounting for over 1.5 million posts across Reddit (Shing et al., 2018). The dataset included many expertly annotated users on a four-level scale of risk of suicide. The task was broken down into three subtasks:

Task A: Assess the level of risk of users based on their posts on SuicideWatch

Task B: Same risk assessment as Task A but using the users' overall posts on Reddit.

Task C: Screening based on users opting into having their social media use monitored

2.2.2 Shared Task 2024

More recently, the CLPsych 2024 Shared Task focused on making use of LLMs for identifying evidence supporting the suicide risk level of individuals on social media (Chim et al., 2024). Similarly to in 2019, the same Reddit dataset was used, with slight variations, with annotated risk levels by post across the SuicideWatch subreddit (Shing et al., 2018; Zirikly et al., 2019). This year's first subtask was to use a model to provide evidence supporting the level of suicide risk of a user based on an individual post, highlighting relevant phrases within the text. The other subtask required the models to generate a summary supporting a user's risk level across multiple posts by the same individual.

2.2.3 Performance metrics

For the Shared Task 2024, Task A was evaluated using the average maximum recall-oriented BERTScore (Zhang et al., 2019). This gave a recall score alongside a precision score by averaging the maximum precision-orientated BERTScore for each predicted evidence span against each gold evidence span. Also, a weighted version of recall is reported, and an F1 score, the harmonic mean between precision and unweighted recall. In task B, a Natural Language Inference (NLI) model is used for summary evaluation (Laurer et al., 2024), measuring consistency based on the absence of contradictory information when compared against a "gold standard" summary. The NLI model calculates the mean probability that each sentence in the summary contradicts the sentences in the "gold standard" summary. Alongside the average consistency score, a contradiction score is evaluated by averaging the maximum contradiction probability of a predicted sentence.

2.3 Model architectures

Within this section, three critical solutions to the CLPsych 2024 shared task are investigated, and their setups are compared. The three setups are as follows:

Solution 1: Alhamed, Ive and Specia (2024)

Solution 2: Tanaka and Fukazawa (2024)

Solution 3: Singh et al. (2024)

2.3.1 Dual Context architectures

Use in CLPsych 2019

One of the groups working on the CLPsych-2019 shared task presented a multi-level dual context setup for assessing suicide risk (Matero et al., 2019), treating suicide-related forum posts separately from other forum content. It combines BERT embeddings for capturing linguistic nuances with user-level assessments such as personality traits and emotional states, aggregated via advanced Recurrent Neural Network (RNN) architectures with attention mechanisms. The dual context BERT architecture performed better than anybody else in the shared tasks A and B, demonstrating the importance of context in language models, particularly in sensitive applications like mental health assessment.

Use in CLPsych 2024

Dual context architectures were adapted within many of the approaches to the 2024 shared task, particularly in Solution 2, utilising the MentaLLaMA-chat-7b framework for generative summaries and a specialised dictionary of suicidal risk words for extractive summaries. With BERT used for extracting highlights, a separate sentiment classifier model fine-tuned on Tweets was used to predict negative sentences (Barbieri et al., 2020). All sentences in the high-risk category were listed as highlights and sorted by the probability of negative sentiment. MentaLLaMA filled in the gap where the highlights were less than 300 words.

The summary for Task B consisted of 4 parts: first, the risk of suicide, followed by a summary based on the number of sentences predicted as the highest suicidal risk by the user. Third, a dictionary-based summary was completed on the suicidal phrases in the negative and suicidal risk sentences. MentaLLaMA gave the remainder of the 300 words from each user's posts.

For the evidence highlight submission, the model achieved the highest precision (0.919) across all teams when using only the suicide risk classification and the highest recall (0.944) when combining sentiment classification and MentaLLaMA. In the summary generation task, it performed subpar with a consistency rating of 0.944. The combination of models within a single architecture is illustrated in Figure 2.1. This solution demonstrates the benefits of combining models for understanding different contexts.

2.3.2 Extraction level

Lexical Extraction

Solution 1 to the CLPsych-2024 shared task utilised the Meta-Llama-2-7B chat LLM, a variant of the LLaMA series, fine-tuned using publicly available instructional datasets including over a million human annotations (Alhamed, Ive and Specia, 2024). As part of this, lexical extraction was used, with previous research showing that lexical features in models can enhance prediction outcomes (AlHamed and AlGwaiz, 2020). Using a predefined list of words associated with suicidality from previous research (AlHamed, Ive and Specia, 2022), and adding depressive words found from manual inspection, the two words before and after the lexicon-found word are used to make a 5-word sentence.

This model provides an overall recall score of 0.577 due to errors in the validity check phase from the previous spell-checking process. For the posts not affected by this issue, a recall score of 0.887 was given, a result on the weaker end of all of the submissions made. In task B,

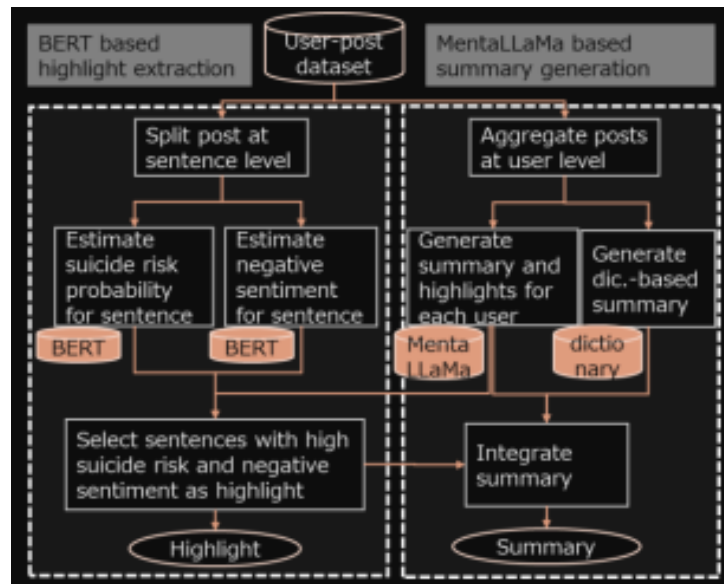


Figure 2.1: Pipeline of architecture (Tanaka and Fukazawa, 2024)

the mean consistency of the model was about average compared to other submissions (0.964), but this Llama model had the smallest max contradiction of 0.060. The authors demonstrated that the use of lexicons with the Llama model achieved notable results, suggesting this as a way forward in the subject. The fundamental limitations were that Llama2 7b was limited to handling only 4096 tokens, resulting in trimmed posts and information loss, as well as a lack of training on specific mental health data.

Sentence-level extraction

With Solution 2 and its dual-context architecture, the BERT model is used to classify sentences for their suicide risk based on the presence of specific phrases linked to suicide. The model is trained with labelled sentences indicating direct expressions of intent. Sentence-level extraction is chosen to reduce irrelevant context without missing crucial words around the important phrase. Training data was prepared from sentences with direct expressions of suicidal intent, ensuring a balanced dataset of suicidal/non-suicidal content to prevent any bias.

The BERT-base classifier was fine-tuned using several steps, including inserting a [CLS] token, serving as an aggregate representation for classification purposes. Once the sentences are tokenised and converted into embeddings, a softmax layer is added to classify sentences into either high or low/no risk based on the embeddings. With a batch size of 8 and a learning rate of $2e-5$, a warmup ratio is used to increase the learning rate in early training, and the model usually converges after about 50 epochs.

2.3.3 Prompting

A vital issue with Solution 2's method was identified in the dictionaries used where sentences related to suicide but without intent introduce noise data. The author recognised that moving forward there is potential use in a more supervised training approach (Sandu, Mihailescu and Nisioi, 2024) and generative LLMs with prompt engineering.

Zero-shot prompting

Solution 1's Meta LLaMA LLM was used in a zero-shot learning configuration, generating responses without additional fine-tuning on data specific to the task. This approach's performance was enhanced using prompt engineering, with the team using specific prompts designed to extract summarisation and highlight evidence directly related to the suicide risk levels from the posts.

Solution 3 outperforms every model on task B of the 2024 shared task. It uses the Mixtral7bx8 LLM (Jiang et al., 2024) with a zero-shot prompting strategy, including meta-information. The zero-shot approach does not rely on task-specific training; instead, it utilises prompts that concentrate on specific aspects and meta-information from multiple user posts. This meta-information involved collecting every post by a user, selecting the most common emotion and sentiment and using it to represent the user's state as a whole.

In the CLPsych shared task, only one paper performed better on task B, with a consistency score of 0.979 and max contradiction of 0.064 (Uluslu, Michail and Clematide, 2024) compared to 0.977 and 0.079, respectively, for this model. Interestingly, the model created by Uluslu's team also implemented a zero-shot prompted Mistral-7b LLM with meta-information, suggesting a slightly better overall setup.

Chain-of-Thought Prompting

Solution 3 gets its best results for Task A using the Tulu-2-DPO-70b model (Iverson et al., 2023), another LLM, alongside chain-of-thought prompting. It gives a higher precision than the MentaLLaMA model mentioned earlier but a slightly lower recall score. Chain-of-thought is used as a few-shot prompting method strategy of in-context learning.

The preparation of this utilises Sequence-BERT (Thakur et al., 2020) and pre-trained RoBERTa-base models, where user posts are represented in vector space with semantic, emotional and sentiment dimensions. The general method involves identifying posts for further analysis, with manual review to narrow down to three to be annotated. Chain-of-thought prompting involves guiding the model in a structured thought process, with the goal of identifying text relevant to emotions, behaviour, and motivation. Despite a strong performance, Solution 3 demonstrates the need for a more inclusive model and diverse prompting strategies, with less reliance on manual annotation.

2.4 Psychiatry and clinical mitigation

Literature generally shows that healthcare workers such as nurses frequently have negative views towards people with mental health issues or those who self-harm (Harmer et al., 2024) and physicians' beliefs about suicide can often include misinformation (Bocquier et al., 2013). These views are an issue in the early detection of suicidal intent or mental health issues, as these professionals are the only trained face-to-face support that people are likely to see. There are various examples of suicidal intent assessments that are used to diagnose and provide people with the support that they need.

2.4.1 Characteristics of intent

The American Psychiatric Association (Silverman et al., 2015) sets out clinical guidelines for assessment as shown in Table 2.1. These characteristics are fundamental in any diagnosis and should always be considered when using Machine Learning models to determine the risk of suicide. While not all of these will be included in every social media post, it is crucial that they are still considered before giving a response to the user.

Table 2.1: Characteristics of suicidal intent (Silverman et al., 2015)

Characteristic	General Considerations
SUICIDAL IDEATION	Thoughts about life and death Control over thoughts Frequency, length and triggers of ideations Active vs Passive ideation
PLAN	Have methods been considered? How detailed is the plan? Has the writer prepared? Has a date/time been chosen?
ACCESS TO MEANS	Does the writer have access to the chosen method? Does the person have career-based knowledge or experience How plausible is the plan
INTENT	What is the primary reason for suicidal thoughts? How intense is the person's desire to die? Has the person made steps to avoid being discovered? Has the person shared their intent with anyone?
LETHALITY	Does the person believe their plan would be lethal? Does the plan include use of drugs or alcohol?
PROTECTIVE FACTORS	Are there any barriers that may prevent the suicide attempt? If the current issues were resolved would the person still want to die? Are there any social factors that could be keeping the person living?
PREVIOUS ATTEMPTS	What was the method and reason for the previous attempt, and when was it? Has the person ever self-harmed?

2.4.2 Mitigation Strategies

With the use of volunteers for crisis intervention becoming so familiar over the past decade (Lester, 2009), it is commonly seen that different cases of suicidal individuals are sent significantly different responses by hotlines (Shahar, 2006). The variation suggests a direct relationship between messages of intent and the strategy with which the volunteer responds. Generally, these associations are seen:

1. Messages with feelings of irreversibility and emptiness are linked to cognitive change inducement and persuasion.

		Response strategies*							
		ES	EGS	EMP	INT	CCI	PER	ADV	REF
Loss of control	Yes	94	39	20	30	41	26	30	4
	No	92	30	22	22	59	27	18	11
Acute loneliness	Yes	90	45	13	33	65	28	23	7
	No	94	28	25	22	46	27	23	10
Emptiness	Yes	89	32	25	32	61	43	29	7
	No	93	34	20	23	50	22	21	10
Narcissistic wound	Yes	96	32	36	32	50	32	18	9
	No	92	34	18	24	53	26	24	9
Irreversibility	Yes	100	46	23	27	41	46	41	5
	No	91	31	21	25	55	22	19	10
Loss of energy	Yes	96	17	30	17	48	17	26	4
	No	92	38	19	27	53	29	22	10
Emotional flooding	Yes	86	21	29	14	50	43	29	7
	No	93	35	20	27	52	25	22	9
Cognitive attribution	Yes	93	38	26	21	71	36	14	5
	No	92	31	18	27	42	22	27	12
Level of risk	Yes	86	43	0	14	14	71	29	0
	No	93	33	22	26	54	24	22	10

*Strategies: ES = Emotional support; OGS = Offering group support; EMP = Empowerment; INT = Interpretation; CCI = Cognitive change inducement; PER = Persuasion; ADV = Advising; REF = Referring.

Figure 2.2: Use of strategies by characteristic of the messages, in percentages. From (Gilat, Tobin and Shahar, 2011)

2. Messages with feelings of frustration are linked to emotional support and the offering of group support methods.
3. Messages that mention self-criticism or self-injury are linked to a strategy of empowerment.
4. Messages that mention helplessness are linked to strategies that focus on emotional and group support.
5. Messages with high suicide risk are linked to strategies of persuasion.

A study was completed on a dataset from an online support forum to investigate further the ideal mitigation strategy based on the direct characteristics of suicidal intent or mental health decline (Gilat, Tobin and Shahar, 2011). The findings highlighted the significant differences between support strategies; emotion-focused and cognitive-focused strategies were used. Table 2.2 shows the findings. As this directly considers the reason for suicidal intent, it suggests that there are uses in AI models providing initial support for those at risk to help the individual get the right help.

2.4.3 GPT in Psychiatry

Despite significant rises in the uses of Large Language Models across healthcare, psychiatric specialists have received limited attention, likely due to ethical and legal limitations of LLMs in this area (Cheng et al., 2023). While Deep Learning has proven beneficial, the use of chatbots such as OpenAI's GPT models has yet to achieve its potential. The GPT-4 model was proven to have excellent cognitive empathy, solving 95% of false-belief tasks (Kosinski, 2024). However, it is recognised that these models perform poorly on personality assessment and average on suicide assessment despite the excellent results in sentiment analysis (Amin, Cambria and Schuller, 2023). This finding demonstrates that the current applications of ChatGPT and similar models are constrained by their setup being that of a chatbot rather than a specialised AI tool and that future LLMs could bridge this gap to support psychotherapy tasks.

2.5 Summary

This chapter discusses the detection of social media users who are at a high risk of suicide and where it has benefited from Natural Language Processing methods over the past decade. This discussion covered the evolution of techniques, such as GloVe embeddings, LSTM layers, and newer transformer-based architectures, such as BERT. Having covered the CLPsych tasks and how performance on them is measured, many different approaches were evaluated. These introduced various LLMs combined with varying approaches with solid advantages and disadvantages.

Dual-Context architectures promised to shape future approaches, allowing different parts of the data to be processed differently. Lexical and sentence-level extraction demonstrated their ability to make the processed data more relevant by not considering other information. Different prompting strategies were shown to combine with LLMs to tackle tasks depending on the nature of both the LLM and the specific task's requirements. Finally, a section on psychiatry was completed, gaining an understanding of appropriate responses for different types of users, depending on the language used in the post that is classified as showing signs of suicidal intent.

Chapter 3

Methods

This chapter captures the overall setup used to meet the aims and objectives set out in Chapter 1. This includes the dataset used, the requirements of the architecture, and the data cleaning methods used. The chapter then discusses the overall setup of the investigation, including how the requirements were met.

3.1 Dataset

A publicly available dataset was obtained from Reddit (Gaur et al., 2019), including the posts of 2181 users across forums (Subreddits) discussing suicide. Five hundred of these users were annotated by four practising psychiatrists using the guidelines of the Columbia Suicide Severity Scale (C-SSRS) (Posner et al., 2011), including a 5-label classification of users:

Supportive: Another additional category for users that discuss suicide with no history of risk.

Indicator: An additional category with risk factors from discussion of the topic, but with no desire for self-harm.

Ideation: The usual lowest level of the C-SSRS, where a user has thoughts of suicide, often due to illness or loss.

Behavior: The C-SSRS rating for a user with ongoing or historical self-harm or who has thoughts of suicide planning.

Attempt: The C-SSRS rating for a user who has taken any action that may result in death.

The dataset removes all personal or identifiable data, resulting in all posts being fully anonymous. It also includes a list of "lexicons"—words common to users at any non-supportive risk level.

3.2 Architecture Requirements

The model must meet various requirements to satisfy the project aims and objectives. As discussed in Chapter 2, incorporating a dual-context involves significant benefits. This task's proposed approach treats user sentences separately from the overall posts, forming a more robust overall model.

Requirements of the sentence level model:

1. Train and incorporate a model to remove any sentences that are not relevant to suicide or mental health.
2. Train and incorporate a model to separate sentences of a supportive nature.
3. Utilise an LLM to assign each sentence a risk rating.
4. Filter to the five most negative sentences, producing "highlights".

Requirements of the post level model:

1. Assign the user a rating based on the risk level of their highlights.
2. Compare LLMs and prompting methods to summarise each user's activity on Reddit.
3. Use meta-information from the sentence model and compare summaries.
4. Use the summary to generate a relevant chatbot response directly.

3.3 Data Cleaning

With the dataset loaded, the posts are split apart from each other and linked with their corresponding userid in a dataframe. The sentences were then cleaned of any non-ASCII characters, with double spaces and other symbols removed. Information inside brackets or quotations was also cleaned out of each sentence, as much of this information would be unuseful or cause noisy data and issues going forward. Finally, the posts could be split out into individual sentences to allow for sentences to be extracted individually; unlike previous research, this was done around complete sentences rather than partial sentences to provide slightly more context.

3.4 Model setup

Figure 3.1 visualises the overall setup of the model, which incorporates six different Machine Learning-based stages to produce quality user highlights, summaries, and chatbot responses. The following sections dissect this.

3.4.1 Sentence-based model

The setup of the dataset was manipulated by setting up simple classification tasks to be evaluated by different models, with the aim of narrowing down to the sentences which are most relevant for the overall task.

In order to meet requirement 1, a BERT model was utilised alongside the list of commonly suicidal words provided by the dataset in order to filter out any sentences that were not related to the topic of suicide or mental health. The model was fine-tuned and applied to the entire list of sentences.

With irrelevant sentences removed, another a similar binary classification approach was taken to requirement 2, with the user labels used in order to train a BERT model on the supportive nature of sentences. By classifying whether a sentence was likely to be supportive or not, a more focused approach could be taken in the later stages.

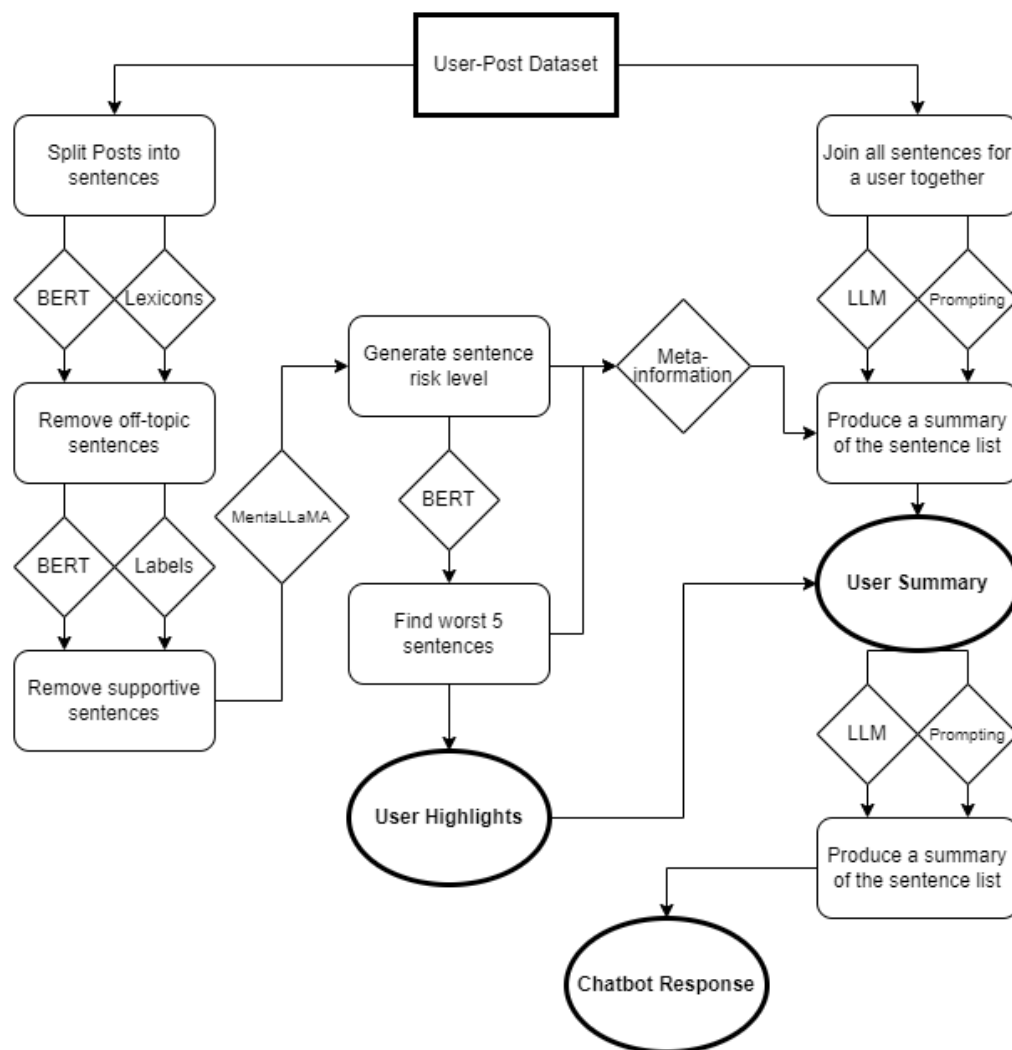


Figure 3.1: Flow Chart of Model Setup

Requirement 3 required using an LLM to generate a risk level for each sentence in the dataset. A zero-shot, direct prompted MentaLLaMA was perfect for this, as the model is pre-trained on significant mental health data and designed for classification tasks. The highest-risk sentences were returned to a BERT-based model to rank sentences by sentiment, producing the highlights for the user, meeting the fourth requirement.

3.4.2 Post-based model

Firstly, an overall rating for the user was generated based on the risk level of the highlighted sentences, and the number of supportive sentences occurring from the user. This satisfied requirement 5.

In order to meet the second requirement of the model focused on overall posts, different LLM and prompting techniques were compared. Meta-information was also provided from the sentence-based model, including the user risk rating and the number of supportive sentences across the user's posts. This meets the third requirement. The performance of the LLMs and prompting strategies, including the effect of meta-information, was evaluated by several metrics, investigating the balance of an informative and accurate model.

To satisfy the fourth requirement, the setup utilises the user summary to produce a chatbot-based response to the user. Again, the LLMs and prompting strategies are compared, including some new ideas, as this task had not previously been tried in this context. Meta-information is excluded as this task is designed to be completed directly from the provided summary.

3.5 Summary

The dataset used has been discussed, the requirements set out for the investigation, and the use of the data has been presented thoroughly. The overall setup and design used to meet the requirements is summarised, demonstrating how each part satisfies the corresponding needs.

Chapter 4

Implementation

This chapter describes in more detail the implementation of the model setup explained in Chapter 3, including the intermediate results of each stage of the architecture.

4.1 Sentence level model

We discussed earlier the benefits of sentence-level extraction, which allows for a better understanding of context. By utilising a combination of Machine Learning techniques, a complete model can be set up that allows for a semantic type analysis of sentences to generate highlights from the text.

4.1.1 Sentence topic prediction

The first task involved removing sentences from the dataframe that are not on the topic of suicide or mental health, as these would potentially cause issues later on.

Setup

As this is a simple binary classification task, a simple BERT base uncased model was selected to identify sentences that are on-topic. However, as the large majority of the dataset itself discusses suicide, the use of a secondary dataset was crucial. This data was obtained from Kaggle (Fekadu, 2019), a dataset of 100,000 random English sentences ending in punctuation, similar to the Reddit dataset. These random sentences were then cleaned and filtered similarly to the suicidality dataset to ensure the sentences were of the same overall format.

To further improve the setup of the task, a sample of the user sentences needed to be extracted that only included sentences directly discussing suicide or mental health issues. To do this, the lists of lexicons gathered earlier (Gaur et al., 2019) were combined, providing a list of common phrases that would likely only appear in a user's post when partaking in discussion on these topics. Therefore, a new dataframe was built that only included sentences that include these phrases and a roughly equal amount of sentences from the Kaggle dataset. The number of each sentence class was intentionally made uneven to prevent the model from trying to force a 50/50 split, requiring it to learn the features of the sentences themselves.

The pre-trained BERT model was initialised with a learning rate of $1e-5$ and utilised learning

rate warmup and decay functions over three total epochs. When applied to all user sentences, those marked as unrelated to suicide or mental health were removed.

Testing

The model learned the features very well, giving an accuracy of 99.5% and an F1 score of 99.6%. The structure of language is likely different in the random sentence dataset to that of our Reddit users, so the model is tested on the remainder of the sentences of the gold standard 500 users dataset. Upon inspection, the model performs well, removing sentences such as "the pain was mostly caused by a nerve which was injured", where the user is discussing pain but not suicide or mental health issues. Removing sentences like this is crucial as they would likely cause false negatives later on. The model also keeps sentences such as "i really don't want to ask for help.", where there are no phrases/lexicons present but it has marked it as potentially discussing mental health.

Table 4.1: Sentence topic prediction model

Epoch	Training Loss	Validation Loss	Accuracy	F1
1	0.0304	0.0479	0.9858	0.9882
2	0.0135	0.0239	0.9947	0.9945
3	0.0014	0.0238	0.9949	0.9957

4.1.2 Supportive sentence prediction

With the end goal to use LLMs to process sentences and posts, it is noted that many of these models deal poorly with sentences supporting others in need, often assigning these as a risk of suicide themselves. This is demonstrated in the approach taken by Tanaka and Fukazawa (2024), where a significant number of "supportive" sentences are identified as evidence of a user's suicide risk, masking proper risk sentences.

Setup

By using the labels provided for users in the gold standard dataset, the list of sentences can be filtered down to only those marked as "Supportive" and "Attempt". With most non-relevant sentences already removed, all of the sentences of "Supportive" users should be of a supportive nature. However, as many of the users marked as "Attempt" will also have sentences supporting others, manual data inspection was required to remove any sentences clearly supporting other users in need.

With a combined dataframe of supportive and clearly non-supportive sentences, a training/test/-validation split is performed on it to provide to a newly initialised BERT model. At this stage, data augmentation techniques were introduced upon 20% of the training set. These techniques include a random selection of backtranslation - where the sentence is translated to French and back again, and synonym replacement - where a random word is replaced with a synonym. Data augmentation greatly supports the model in learning different types of English language and provides more total training data. The list of sentences is then reduced to a roughly even size between supportive and non-supportive data.

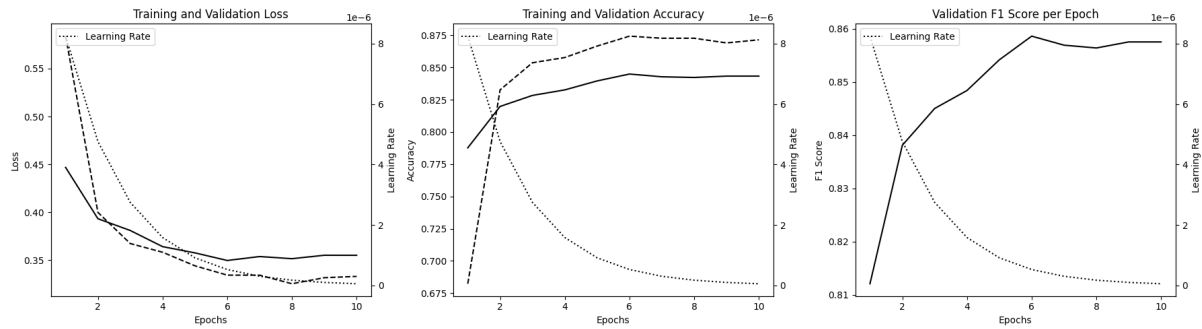


Figure 4.1: Loss and Accuracy for Training and Validation set, and F1 Score for Validation Set

The model used is the pre-tuned BERT base uncased again, but this time with slight changes to its architecture. Firstly, a dropout layer is added to the model, increasing the model's ability to generalise to new data by preventing situations where the model tries to adapt to correct previous mistakes. L2 regularisation is also introduced, encouraging the model to distribute weights across all features evenly. The introduction of both of these techniques helps to prevent overfitting to the training set, and with the uneven ratio of classes. A callback function is also introduced, where the model will roll back to its previous setup if the loss of the function increases. The model is trained over ten epochs with a learning rate of $1e-5$, and rate warmup and decay are used as before.

Results

Once trained, the model gives an accuracy on a new test set of 0.845 and an F1 score of 0.859. Figure 4.1 shows the loss, accuracy and F1 score on the training and validation sets across the epochs. The model learns quickly and well and converges after about six epochs. While some overfitting is present, as demonstrated by the lower training loss, it is minimised due to the model's sophisticated setup. While the accuracy is not exceptionally high, with some testing, it is as high as it gets without extreme overfitting. Also, less than 85% accuracy does not necessarily mean that the model is not learning the features well.

Investigation

A confusion matrix is also produced, as shown in Figure 4.2, allowing visualisation of the number of sentences in each class that are predicted differently from their actual value. The result is positive, as the "Supportive" class has a higher percentage of sentences predicted differently than the set without a manual inspection. The "Attempt" class had its sentences inspected and, therefore, had a lower number predicted as supportive. With this model applied to all the user's sentences, it was seen to correctly pick up language regarding others' issues and mark it as supportive. It also correctly reclassifies some "Supportive" users' sentences as non-supportive when the sentence is irrelevant but was missed by the topic prediction model.

4.1.3 Sentence classification

By removing all sentences for a user that are relevant and non-supportive, the remainder of the data is likely sentences that are indicators of direct risk of suicide. In order to support the summary for a user, each sentence can be ranked in terms of its inherent risk of suicide. By taking the Columbia Suicide Severity Scale mentioned earlier, 5 "risk" levels can be defined:

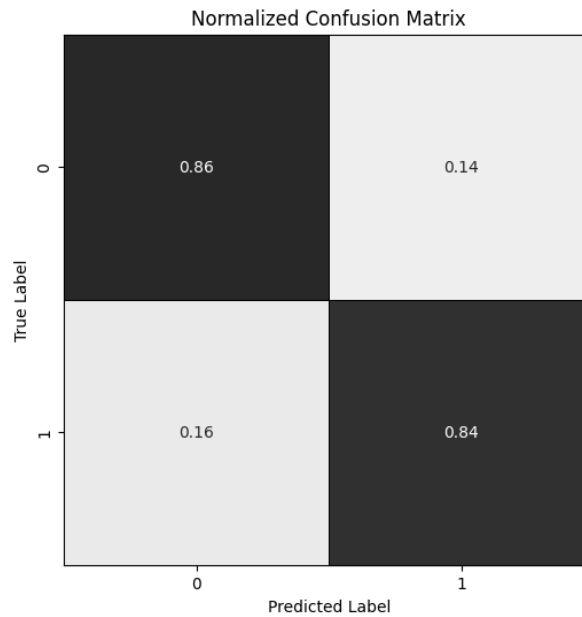


Figure 4.2: Confusion Matrix for Test Set

Very Low: The patient is either supporting other patients or does not mention any ongoing distress or negative feelings.

Low: The patient mentions ongoing distress or negative feelings, including past trauma, such as the suicide of others.

Medium: The patient mentions thoughts of personally wanting to die or hurt themselves but without any specific plan.

High: The patient mentions actions taken towards a potential suicide attempt or mentions ongoing self-harm.

Very High: The patient mentions recent actual suicide attempts, future planned attempts, or severe self-harm with a clear intent to die.

By including a "Very Low" category, any sentences that were missed previously and are likely of a supportive or irrelevant nature can be excluded from the sentences carried forward from this point.

Setup

In order to complete this task, MentaLLaMA is chosen as it is proven to be strong at mental health-related classification tasks (Tanaka and Fukazawa, 2024), especially with a zero-shot, direct prompting approach, without any fine-tuning. At this point, few-shot and chain-of-thought prompting are not utilised or compared as it would be against the nature of the task (Kundu, 2023). After some testing, a prompt was selected that would produce a result that was similar enough for each sentence, making it much easier to convert the output of the LLM into a specific sentence label. The prompt used is shown in Figure 4.3, with examples shown in Appendix A. The output of the LLM is used to assign each remaining sentence its risk level number (1-5).

```
[INST]
Please clarify the following sentence into one of the following categories:

1: Patient is either supporting other patients, or does not mention any ongoing distress or negative feelings.
2: Patient mentions ongoing distress or negative feelings, including past trauma such as suicide of others.
3: Patient mentions thoughts of personally wanting to die or hurt themselves but without any specific plan.
4: Patient mentions actions taken towards a potential suicide attempt or mentions ongoing self-harm.
5: Patient mentions recent actual suicide attempts, future planned attempts, or severe self-harm with a clear intent to die.

Only place the sentence in a higher category if it is clear that it belongs there.
Sentence: "{sentence}"

Category:
[/INST]
```

Figure 4.3: Prompt used for MentaLLaMA classification task

4.1.4 Highlight Generation

In order to produce highlights for a user, the five sentences labelled as highest risk were selected by choosing sentences from the highest risk group down to the lowest. If there are more sentences within a risk level than there are left to be selected, all the sentences within that risk level will be ranked according to their negative sentiment. For this, the BERT negative sentiment model by Barbieri et al. (2020) is selected, as it requires no training and is explicitly pre-trained on mental health data from Twitter. The sentences ranked as the most negative are used to fill the rest of the highlights.

4.1.5 Stage testing

To test the quality of the highlights generated at this stage, a BERTScore model is incorporated (Zhang et al., 2020), allowing for a quantifiable value for how well the highlighted sentences represent the user's overall posts. This gave:

Precision: 0.871

Recall: 0.820

F1 Score: 0.845

While this result is relatively ambiguous, it demonstrates that the extracted highlights do, in fact, represent the user well on average and are sufficient to be passed on to the post level model. The highlights are provided to the next parts of the model, alongside the five sentence level ratings, and the user's overall ratio of supportive sentences.

4.2 Post level model

Utilising LLMs for overall summary benefits from aggregating all user posts as there is a broader context that is important. However, the sentence level model plays a part in providing meta-information to the model to help with processing.

4.2.1 User risk rating

By using the five sentences collected as highlights, a risk rating is generated for each user. These risk ratings are as follows:

No Risk The user does not have any highlights (all sentences are supportive)

Very Low Risk Low Risk users, but with the significant majority of the user's overall sentences supportive.

Low Risk The user averages a risk rating of "2" in their highlights

Medium Risk The user averages a risk rating of "3" in their highlights

High Risk The user averages a risk rating of "4" in their highlights

Very High Risk The user averages a risk rating of "5" in their highlights

Further rules must be followed in order to be placed into a category. For example, to avoid under-rating a user, if they have a sentence that is ranked 4 or 5, the user must be placed in Medium Risk or higher. However, to avoid a badly classified sentence skewing non-suicidal users, this rule will be ignored if the remainder of the highlights are low risk. The selected code for this is listed in Appendix E.1.

4.2.2 Models

Tulu-2-DPO models have already been proven to be strong in summary generation tasks when provided with meta-information (Singh et al., 2024). Despite the fact that this model usually benefits from Chain-of-thought-based prompting, the previous approaches used zero-shot prompting and based their focus on the quality of meta-information. Mistral7b models have also been demonstrated to be strong in zero-shot settings with meta-information (Uluslu, Michail and Clematide, 2024). For this investigation, only the Mistral-7b-instruct-v0.3 and Tulu-2-DPO-7b models are compared.

4.2.3 Prompting methods

Zero-shot and Few-shot prompting are commonly compared and utilised with LLMs in many different tasks, but zero-shot has consistently performed better on this task in the past. Chain-of-thought prompting is not commonly used for this task as it is naturally used in a few-shot setting. For this investigation, Chain-of-thought prompting was tested in its ability to support the zero-shot method by providing the models with clear step-by-step instructions.

4.2.4 Use of meta-information

Meta-information was provided from the sentence-based model. The data included the risk rating of the user and their ratio of supportive sentences. This was provided to support the LLM in understanding the level of risk being analysed, and the supportive nature of users. The meta-information was provided to the models within prompts.

4.2.5 Summary generation

As mentioned above, summaries were produced for each selected model using different prompting methods and meta-information. Certain combinations were not used if they produced sub-par results in the intermediate investigation stage to focus the investigation.

Primarily, each model was utilised in zero-shot settings, but with the inclusion of chain-of-thought prompting, and meta-information. A general example of these prompts and

```

In one paragraph, summarise the provided text from the author on Reddit:

Full input text:
{Posts}

Meta-information:
{Meta-information}

Summary:

```

Figure 4.4: Zero-Shot prompting method, with optional meta-information

```

As a mental health expert, your task is to summarise in one paragraph, directly
from the provided input from the author on Reddit. When summarising, consier these aspects:

Emotions: Evaluate emotions, from sadness to intense psychological pain.
Cognitions: Explore thoughts and perceptions about suicide, including level of suicidal thoughts.
Behavior: Evaluate actions related to suicide, such as access to means or set plans.
Motivations: Consider the motivations behind their desire to die.
Interpersonal and Social Support: Investigate social support and relationships, understand feelings toward others.
Mental Health-Related Issues: Consider psychiatric diagnoses such as schizophrenia, bipolar, anxiety, eating disorder.
Additional Risk Factors: Consider socioeconomic factors, suicide behavior by others, chronic medical conditions, etc.
Supportive Nature: Take into account authors who are supporting other users in their writing.

Full input text:
{Posts}

Meta-information:
{Meta-information}

Summary:

```

Figure 4.5: Chain-of-thought prompting method, with optional meta-information

meta-information used for each model are shown in Figures 4.4 and 4.5, motivated by the approach used by Singh et al. (2024). Complete examples are shown in Appendix A.

4.2.6 Chatbot response

For the chatbot response task, each of the LLMs mentioned previously were compared again, using and comparing standard zero-shot prompting with the chain-of-thought alternative. No meta-information was included as this task is designed to be completed directly from the user summary without any additional information.

The prompting was focused by providing both with the methods of seeking help in the UK, and a line explaining that the patient is referred to as "the author". This explanation is crucial as the LLMs are built to respond to all tasks, and by referring to the user as an author, the language models decode this as an instruction, creating their own suicidal messages. Also, by providing the resources available in the UK, it can be tested as if it were in a real-life scenario, where the user would be provided with options to get help.

The chain-of-thought prompting here focused on the mitigation strategies mentioned earlier in Figure 2.2, with the approach taken by the model needing to be adjusted to match the key emotions and characteristics in the original post. Figures 4.6 and 4.7 show a general example of each of the prompts used, motivated by the response strategies mentioned earlier (Gilat, Tobin and Shahar, 2011). Appendix A once again provides full examples of these prompts.


```

In one paragraph, produce a response to the patient (referred to as the author), based on the provided summary.

Assume the patient is based in the UK with the following helplines:

Emergency Services: 999.
Samaritans helpline: 116 123.
SHOUT texting helpline: 85258.

Input summary to be used for response: {summary}

Response:

```

Figure 4.6: Zero-Shot prompting method

```

As a mental health expert, your task is to provide a supportative message to the patient (referred to as the author),
directly from the provided summary.

Consider:
**
Consider these suggested response strategies by potential characteristics of the message, only include the
response strategies most related to the patients's needs:

Loss of control: Advising, Interpretation, Offering Group Support, Emotional Support.
Acute loneliness: Offering Group Support, Cognitive Change Inducement, Interpretation.
Emptiness: Persuasion, Interpretation, Cognitive Change Inducement, Advising.
Narcissistic wound: Empowerment, Interpretation, Persuasion, Emotional Support.
Irreversibility: Emotional Support, Advising, Persuasion, Offering Group Support.
Loss of energy: Empowerment, Emotional Support, Advising.
Emotional flooding: Persuasion, Empowerment, Advising.
Cognitive attribution: Cognitive change inducement, Persuasion, Empowerment.
Level of risk: Persuasion, Offering group support, Advising.

Also, referring the customer to get further help may be useful in extreme cases.
For this task, all referrals must go to UK helplines:

For urgent risk cases - Emergency Services: 999.
For high risk cases - Samaritans: 116 123.
For high risk cases where the patient might not want to call - Text SHOUT: 85258.
**

Input summary to be used for response: {summary}

Response:

```

Figure 4.7: Chain-of-thought prompting method

4.3 Testing LLMs

4.3.1 Summary Testing

With each combination of LLM, prompting method and meta-information trialled, results were generated using a combination of methods. BERTScore was used similarly to its setup in the highlight extraction task, with the precision score calculated.

However, as the summaries include little information directly from the posts, a more sophisticated method is required in order to test the strength of the summary itself. An Information Retrieval (IR) score was generated using an Okapi BM25 model (Robertson and Zaragoza, 2009), ranking the relevance of the summary to the original text. This score checks if essential words and phrases are present in the summary. Also, a Question Answering (QA) model was set up using QAEval (Deutsch, Bedrax-Weiss and Roth, 2021), evaluating how well a summary answers questions about the original text. This QA model gives a score quantifying how informative the summary is and how well it captures the meaning and content of the posts.

4.3.2 Response Testing

For each large language model, the two different prompting methods were compared, and once again, results were generated using different methods. Firstly, BERTScore was used once more to give a precision score. However, similarly to the summary task, the responses include little information directly from the posts. For this specific task, four additional metrics are used.

Firstly, a Flesch-Kincaid reading score is generated, a numeric score between 1 and 100, correlating to how easy the passage of text is to read (Readable, 2024). Alongside this, the Flesch-Kincaid grade level is provided, correlating to the reading grade of the text based on the US grade levels. A higher grade is not better here, as that would make the text more complicated for a less advanced reader to understand.

Additionally, a sentiment score was produced using the same RoBERTa-based model as earlier in this investigation (Barbieri et al., 2020). A similar setup was then incorporated to quantify the empathy score of the response using another RoBERTa-based model (Tafreshi et al., 2021). Both of these values are crucial as they are essential features of any professional response to a patient in need.

4.3.3 Summary

This chapter discusses the in-detail approach taken to setting up, adjusting and testing the architecture. The testing and results for each stage were presented, confirming that the setup was robust and could be utilised to test the different LLMs and prompting methods. With this completed, all data was processed, generating results across all 500 users of the dataset.

Chapter 5

Results

In this chapter, the results of the overall model will be analysed and compared. The sentence based approach had already been tested, with the results produced in Chapter 4, allowing for a focus on the key summary and response generation.

5.1 Summary generation

Tulu-2-DPO-7b generated some interesting results, as shown in Table 5.1. When utilising zero-shot prompting, the pre-trained setup of the model performed very well, with the highest precision and Information Retrieval (IR) score. When meta-information was introduced, the precision of the model slightly decreased, with the IR Score also dropping. However, the Question Answering (QA) score dramatically increased. This increase suggests that by introducing meta-information, the summary becomes more informative but loses some of the relevant phrases from the original text.

Chain-of-thought prompting, while generally improving the ability of models like Tulu-2-DPO-7b, earlier research showed that this is not the case when it comes to summarisation tasks. This is verified in the results, where there is significant information loss when moving to this prompting method. Furthermore, chain-of-thought prompting reduces the QA score, suggesting that the summary also becomes less informative. This is likely due to the way summaries are produced, reducing the quality of text in order to "think step by step".

Table 5.1: Results for Tulu-2-DPO-7b

Prompting	Meta-Information?	Precision	IR Score	QA Score
Zero-Shot Prompting	No	0.8557	18.29	0.165
Zero-Shot Prompting	Yes	0.8526	17.22	0.189
CoT Prompting	No	0.8434	17.76	0.136
CoT Prompting	Yes	0.8429	16.6	0.185

Mistral-7b-instruct-v0.3 produced very similar results to Tulu-2-DPO-7b, as shown in Table 5.2, with the precision matching very closely in the zero-shot setting. However, a much more significant decrease occurs when chain-of-thought prompting is introduced. When inspecting the IR scores, there is a crucial difference to the previous model, with both meta-information and chain-of-thought prompting greatly increase the score. This suggests that the Mistral

model is much more capable of processing meta-information and instructional prompting without losing excess information. Similarly to the previous model, however, the QA score increases with meta-information but generally decreases with chain-of-thought prompting.

Table 5.2: Results for Mistral-7b-instruct-v0.3

Prompting	Meta-Information?	Precision	IR Score	QA Score
Zero-Shot Prompting	No	0.8527	17.33	0.126
Zero-Shot Prompting	Yes	0.8476	18.17	0.179
CoT Prompting	No	0.8398	18.53	0.128
CoT Prompting	Yes	0.8383	18.79	0.144

With the two models compared, it is clear that the Tulu-2-DPO-7b model obtains a more robust result on each of the metrics; however, the Mistral-7b-instruct model demonstrates the strongest overall solution. Where this model can better deal with meta-information, it can produce a more informative model without losing too much of the original information. However, it is evident that the step-by-step process of the chain-of-thought prompting method does not benefit the model and that zero-prompting would be the stronger method.

As the dataset is very different from similar tasks, such as the CLPsych 2024 Shared Task, and the setup is very different from its solutions, it is tough to compare the model against others. However, some examples can be studied and critiqued against our recommended criteria.

An example of a user with the "Supportive" label is shown in Figure 5.1, with the complete list of their posts and the generated summary. At first glance, the model has clearly covered all the information in the posts and well summarised the feelings from the perspective of the Reddit user. However, on further inspection, it can be seen that the model has performed excellently on inferred reasoning, recognising the type of language used and what this suggests about the user. Without the meta-information included in the prompt, the model fails to recognise much of the modern language utilised that suggests the level of suicidality or supportiveness of the user.

Due to the effectiveness of the zero-prompted Mistral7b setup with meta-information, the approach is extended to produce summaries of all 500 users, to be used in the response generation testing.

5.2 Response generation

The results of both Tulu-2-DPO-7b and Mistral-7b-instruct-v0.3 are shown in Table 5.3, with both models performing well across the average user, providing the patient with structured support and hotline links.

When chain-of-thought prompting was introduced to the Tulu-2-DPO model, the readability of the response and BERTScore precision decreased slightly. However, the sentiment of the post was much more positive. The empathy score had a slight decrease as well. The decreasing precision and readability are of no surprise as we gave the model a prompt introducing many technical words and phrases. The sentiment of the post becoming more positive demonstrates the ability of chain-of-thought prompting with this model, with it better recognising users in need of emotional support by thinking step-by-step.

Posts:

Its not a viable option, and youll be leaving your wife behind. Youd Pain her beyond comprehension. It sucks worrying about money, I know that first hand. It can definitely feel hopeless, as you seem to be very aware of. Your wife might need to chip in financially. I know time is an issue, but even 10-15 hours a week could alleviate a lot of the pressure. In the meantime, get your shit together - write that resume tomorrow. No excuses, get it done and send it out. Whether you believe in some sort of powerful being or force governing things or not, things really do work themselves out. This is a big test for you, and youll pull through. Just try to stay as positive as you can and everything will work out.']

Summary:

The author has no suicide risk. 91% of the author's sentences support other users. The author empathizes with the reader's financial struggles and the associated stress, acknowledging it can feel overwhelming. They suggest seeking part-time work to alleviate some financial pressure, even if time is limited. The author encourages the reader to write their resume the next day and send it out without delay. They also express confidence in the reader's ability to overcome the current challenges, emphasizing the importance of maintaining a positive outlook. The author concludes by suggesting that things will work out, implying a belief in a positive outcome, regardless of personal beliefs in a higher power or destiny.

Figure 5.1: An example of a Supportive User's posts and the generated summary

Table 5.3: Model response generation results, with Tulu-2-DPO-7b abbreviated to Tulu

Model	Prompting	Precision	Readability	Grade	Sentiment	Empathy
Tulu	Zero-shot	0.846	64.3	9.13	0.664	0.9922
Tulu	CoT	0.840	63.3	9.23	0.767	0.9920
Mistral	Zero-shot	0.838	66.1	7.94	0.759	0.9922
Mistral	CoT	0.833	69.0	7.77	0.745	0.9922

The Mistral model's scores were very interesting, producing a much better readability overall than the Tulu-2-DPO model, with a lower school grade needed to understand the message. This result is very positive as it means those with worse reading and comprehension skills are still able to receive the same level of support as others. The Mistral models had a slightly better readability when chain-of-thought prompting was introduced, with sentiment scores slightly lower than that of Tulu-2-DPO-7b's CoT result. Interestingly, when the posts are filtered down to those with a "Medium" or higher suicide risk, the readability goes up even more when using the Mistral model. However, the sentiment score drops significantly.

This decrease is likely due to the complicated nature of the summaries, where users have a combination of sentences discussing their own risk and those supporting other users. In fact, with some manual review, the best results are achieved when a user has very few supportive sentences and is at high risk, precisely what would be needed for this task. Chain-of-thought does, however, mitigate this, resulting in slightly stronger results on high-risk users who are also heavily supportive of others. Appendix C has an example of responses generated; on manual review, it can be seen that chain-of-thought prompting seems to have a significant impact on the mitigation strategies taken by the LLM.

5.3 Discussion

The dual-context approach taken across this investigation produced clear and promising results by combining models and ideas.

5.3.1 Sentence-based model

The sentence level model was designed to produce relevant highlights from the full user posts, as a list of the five highest-risk sentences. Off-topic sentences were correctly removed, removing around 10% of all sentences by utilising a simple BERT classifier. The 99.5% accuracy score is excellent, and the use of a secondary dataset demonstrated to be very beneficial in this task, to reduce manual annotation. This met the first of the requirements set out in Chapter 3.

Supportive sentences were then predicted by utilising the most supportive sentences and the sentences of a subset of the highest-risk users. Some manual annotation was required here due to the lack of information captured in the original dataset. Various Machine Learning approaches were incorporated here, with various layers applied to the base BERT model to reduce overfitting. An accuracy of 84.5% was achieved demonstrating a robust setup and a great method of removing supportive sentences from further investigation. This satisfied the second requirement of the sentence level model.

MentaLLaMA demonstrated its excellent ability in mental health related tasks without additional fine-tuning, by allocating each sentence a risk level, to meet requirement 3. While the benefits of MentaLLaMA are already recognised, the model was very useful in removing further supportive or off-topic sentences not picked up in the previous stages.

The fourth requirement was fulfilled through picking the sentences with the highest suicide risk, and utilising a sentiment model as a "tie-breaker" to ensure that the highlights really are the five sentences with the highest suicide risk. The BERTScore precision score of 87.1% was really promising as it demonstrates these five sentences alone cover the majority of the crucial information from the overall posts.

Combining several sentence-based models was really effective, with a semantic-like analysis of the user's overall sentences showing very positive results in its use in highlight extraction and meta-information. The work here substantially helped the summary generation model by providing it with transparent and clear meta-information. The highlighted evidence can then be used alongside a summary to provide evidence for a user's risk.

5.3.2 Post-based model

A 6-level rating produced for each user by leveraging sentence risk levels and the user's ratio of supportive sentences, meeting the first requirement of this model. This risk rating was utilised to prompt the summary generation LLMs better.

Having compared the Tulu-2-DPO-7b and Mistral-7b-instruct models, it is clear that the Mistral model performs significantly better as a whole on text summarisation and response generation tasks. Although Tulu-2-DPO-7b is recognised as a strong performer for the text summarisation task, the Mistral model's ability to utilise the meta-information provided to produce more informative results without losing information is far superior. By utilising the sentence-level architecture in order to generate the meta-information, much more context is provided to the LLM, allowing it to pick up on linguistic nuances that it may have otherwise missed.

Chain-of-thought prompting was proven to produce worse results for both models in the summarisation task; this was unexpected for the Tulu-2-DPO model but is likely due to the task being straightforward, and it reduces the text quality in the step-by-step approach. This satisfied the second and third requirement of the model.

A response to the user was generated for each model and prompting method, as set out in requirement four. Chain-of-thought prompting was proven to be beneficial in the response generation task, allowing the Large Language Models to better recognise users in need of support and the mitigation strategy that would best fit the patient's needs.

The Mistral model was shown to produce much more readable responses in this chatbot-style task, likely due to the efficient setup of the model and focused pre-training. Chain-of-thought prompting did not significantly change the evaluation metrics, although slightly improving the readability of the task. However, on manual inspection, it can be seen that high-risk users with many supportive sentences are processed much better by the chain-of-thought prompting, with the model likely struggling without this extra context.

The summary itself, therefore, significantly impacts this task, and it is likely that the models could benefit from a summary generated on just the suicidal messages within the post. However, this introduces a challenge in current LLMs, which sometimes need help to identify the sentences supporting other users, as shown in the improvement in summaries when meta-information was introduced.

Chapter 6

Conclusion

6.1 Main Findings

This study demonstrates the use and significance of Machine Learning methods and Large Language Models (LLMs) in suicide risk analysis in social media. The usefulness of a dual-context approach was demonstrated. Highlights were successfully generated from user posts through a combination of models, extracting information at the sentence level.

A summary of a user's posts was then produced through different LLMs and extraction techniques, processing data at the post level and using the extracted highlights from the sentence-based approach. Chain-of-thought and zero-shot prompting strategies were trialled on LLMs, with the inclusion of meta-information evaluated, achieving strong performance and identifying more robust combinations.

These models and prompting strategies were used again when leveraging the generated summary to produce a "chatbot-style" response to the user. This demonstrated the efficacy of the LLMs and highlighted the impact of chain-of-thought prompting in this task, with the model using effective mitigation strategies. The research in this response generation is promising, leading the way in advancing computational psychiatric care to reduce the load on healthcare professionals.

In-depth evaluation metrics were utilised to compare the different setups and their ability to retain information while producing an informative summary. This was expanded to the response generation task by utilising metrics to quantify readability, sentiment and empathy of responses.

The findings of this study should help innovate future changes in mental health intervention, identifying users at risk and bridging the gap between the patient and a professional. Moreover, this should result in people struggling with mental health or suicidal thoughts getting the help they need.

6.2 Limitations

This study has a few limitations in its setup. Firstly, only two models are evaluated, Mistral-7b-instruct-v0.3 and Tulu-2-DPO-7B, with other Large Language Models potentially achieving better results. The versions of these models with only 7 Billion parameters were chosen due to

their smaller size, with the larger models requiring significantly more computational power. Additionally, minimal variations of chain-of-thought prompting were used, with other phrasing and methods needing to be investigated thoroughly.

Also, the dataset used potentially needed to be more diverse, with the data solely sourced from one subreddit, and is potentially slightly outdated with the increase of modern language. The model could be tested on other social media websites to test the cross-compatibility of the model with other platforms having a very different format.

The dataset lacked gold-standard highlights, summaries, and responses, so the evaluation metrics used likely did not fully demonstrate the better solutions. This expert annotation would allow for further evaluation of solutions, with the model likely improving.

6.3 Future work

In future research, the response generation task could be developed further by experimenting with more prompting methods and combining them with other summary production methods. This setup could then be incorporated into a medical setting and tested alongside professional supervision to evaluate its effectiveness. With more diverse training data, combining information sources, the model's efficacy would likely improve further and produce more robust results

With higher computational power, more complex LLMs could be utilised and fine-tuned to generate consistent high quality responses to the user. Additionally, by producing more consistent summary styles, the model would likely be much more effective at generating responses to a user.

6.4 Personal Reflection

The work above has been challenging yet rewarding to complete, helping work towards a future where mental health is dealt with more efficiently, and suicide rates are reduced. An important topic to me, I hope to partake in future research in this field, and drive positive change in mental health and suicide mitigation.

Number of words until this point, excluding front matter: 9773.

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Appendix A

Prompting examples

Below is one example of each prompting method for the post summarisation task, and one example of each prompting method for the response generation task.

In one paragraph, summarise the provided text from the author on Reddit:

Full input text:

a book is usually what i do when im getting down, but it doesnt work when i start getting panicky. ill try the carbs, the caffeine doesnt work because ive gotten it in a movie theater and had a soda with me... a few reasons. i feel backed into a corner mostly. and im tired of being tired of everything. if that makes sense. thank you! i understand its a sad thing. but i also want people to realize that there can be humor in anything and its the best way to deal with this. its how i would do it. i really dont want to ask for help. id rather not let anyone know im having these kind of issues.

Summary:

Figure A.1: Zero-shot prompted summary generation

In one paragraph, summarise the provided text from the author on Reddit:

Full input text:

a book is usually what i do when im getting down, but it doesnt work when i start getting panicky. ill try the carbs, the caffeine doesnt work because ive gotten it in a movie theater and had a soda with me... a few reasons. i feel backed into a corner mostly. and im tired of being tired of everything. if that makes sense. thank you! i understand its a sad thing. but i also want people to realize that there can be humor in anything and its the best way to deal with this. its how i would do it. i really dont want to ask for help. id rather not let anyone know im having these kind of issues.

Meta-Information:

The author indicates Low suicidal risk, with 28.57% of their sentences supporting other users.

Summary:

Figure A.2: Zero-shot prompted summary generation with meta-information

As a mental health expert, your task is to summarise in one paragraph, directly from the provided input from the author on Reddit. When summarising, consider these aspects:

Emotions: Evaluate expressed emotions, from sadness to intense psychological pain.

Cognitions: Explore the individual's thoughts and perceptions about suicide, including the level and frequency of suicidal thoughts, intentions of suicide, and any existing plans.

Behavior and Motivation: Evaluate the user's actions related to suicide, such as access to means and concrete plans. Consider their ability to handle difficult/stressful situations and the motivations behind their desire to die.

Interpersonal and Social Support: Investigate the individual's social support or stable relationships, and understand their feelings toward significant others.

Mental Health-Related Issues: Consider psychiatric diagnoses associated with suicide such as schizophrenia, bipolar, anxiety, eating disorder, previous suicidal attempts, and others.

Additional Risk Factors: Consider other factors like socioeconomic and demographic factors, exposure to suicide behavior by others, chronic medical conditions, etc.

Supportive Nature: Take into account authors who are supporting other users in their writing.

Full input text:

a book is usually what i do when im getting down, but it doesnt work when i start getting panicky. ill try the carbs, the caffeine doesnt work because ive gotten it in a movie theater and had a soda with me... a few reasons. i feel backed into a corner mostly. and im tired of being tired of everything. if that makes sense. thank you! i understand its a sad thing. but i also want people to realize that there can be humor in anything and its the best way to deal with this. its how i would do it. i really dont want to ask for help. id rather not let anyone know im having these kind of issues.

Summary:

Figure A.3: Chain-of-thought prompted summary generation

As a mental health expert, your task is to summarise in one paragraph, directly from the provided input from the author on Reddit. When summarising, consider these aspects:

Emotions: Evaluate expressed emotions, from sadness to intense psychological pain.
 Cognitions: Explore the individual's thoughts and perceptions about suicide, including the level and frequency of suicidal thoughts, intentions of suicide, and any existing plans.
 Behavior and Motivation: Evaluate the user's actions related to suicide, such as access to means and concrete plans. Consider their ability to handle difficult/stressful situations and the motivations behind their desire to die.
 Interpersonal and Social Support: Investigate the individual's social support or stable relationships, and understand their feelings toward significant others.
 Mental Health-Related Issues: Consider psychiatric diagnoses associated with suicide such as schizophrenia, bipolar, anxiety, eating disorder, previous suicidal attempts, and others.
 Additional Risk Factors: Consider other factors like socioeconomic and demographic factors, exposure to suicide behavior by others, chronic medical conditions, etc.
 Supportive Nature: Take into account authors who are supporting other users in their writing.

Full input text:

a book is usually what i do when im getting down, but it doesnt work when i start getting panicky. ill try the carbs, the caffeine doesnt work because ive gotten it in a movie theater and had a soda with me... a few reasons. i feel backed into a corner mostly. and im tired of being tired of everything. if that makes sense. thank you! i understand its a sad thing. but i also want people to realize that there can be humor in anything and its the best way to deal with this. its how i would do it. i really dont want to ask for help. id rather not let anyone know im having these kind of issues.

Meta-Information:

The author indicates Low suicidal risk, with 28.57% of their sentences supporting other users.

Summary:

Figure A.4: Chain-of-thought prompted summary generation with meta-information

In one paragraph, produce a response to the patient (referred to as the author), based on the provided summary.

Assume the patient is based in the UK with the following helplines:

Emergency Services: 999.
 Samaritans helpline: 116 123.
 SHOUT texting helpline: 85258.

Input summary to be used for response:

The author empathizes with the reader's financial struggles and the associated stress, acknowledging it can feel overwhelming. They suggest seeking part-time work to alleviate some financial pressure, even if time is limited. The author encourages the reader to write their resume the next day and send it out without delay. They also express confidence in the reader's ability to overcome the current challenges, emphasizing the importance of maintaining a positive outlook. The author concludes by suggesting that things will work out, implying a belief in a self-correcting universe or personal resilience. The author does not indicate any suicidal risk, and their sentences generally support and encourage the reader.

Response:

Figure A.5: Zero-shot prompted response generation

As a mental health expert, your task is to provide a positive, empathetic and supportative message to the patient (referred to as the author), directly from the provided summary.

Consider:

Consider these suggested response strategies by potential characteristics of the message, only include the response strategies most related to the patients's needs:

Loss of control: Advising, Interpretation, Offering Group Support, Emotional Support.
 Acute loneliness: Offering Group Support, Cognitive Change Inducement, Interpretation.
 Emptiness: Persuasion, Interpretation, Cognitive Change Inducement, Advising.
 Narcissistic wound: Empowerment, Interpretation, Persuasion, Emotional Support.
 Irreversibility: Emotional Support, Advising, Persuasion, Offering Group Support.
 Loss of energy: Empowerment, Emotional Support, Advising.
 Emotional flooding: Persuasion, Empowerment, Advising.
 Cognitive attribution: Cognitive change inducement, Persuasion, Empowerment.
 Level of risk: Persuasion, Offering group support, Advising.

Also, referring the customer to get further help may be useful in extreme cases.
 For this task, all referrals must go to UK helplines:

For urgent risk cases - Emergency Services: 999.
 For high risk cases - Samaritans: 116 123.
 For high risk cases where the patient might not want to call - Text SHOUT: 85258.

Input summary to be used for response:

The author empathizes with the reader's financial struggles and the associated stress, acknowledging it can feel overwhelming. They suggest seeking part-time work to alleviate some financial pressure, even if time is limited. The author encourages the reader to write their resume the next day and send it out without delay. They also express confidence in the reader's ability to overcome the current challenges, emphasizing the importance of maintaining a positive outlook. The author concludes by suggesting that things will work out, implying a belief in a self-correcting universe or personal resilience. The author does not indicate any suicidal risk, and their sentences generally support and encourage the reader.

Response:

Figure A.6: Chain-of-thought prompted response generation

Appendix B

Summary examples

User 1:

The author empathizes with the reader's financial struggles and the associated stress, acknowledging it can feel overwhelming. They suggest seeking part-time work to alleviate some financial pressure, even if time is limited. The author encourages the reader to write their resume the next day and send it out without delay. They also express confidence in the reader's ability to overcome the current challenges, emphasizing the importance of maintaining a positive outlook. The author concludes by suggesting that things will work out, implying a belief in a self-correcting universe or personal resilience. The author does not indicate any suicidal risk, and their sentences generally support and encourage the reader.

User 2:

The author on Reddit provides a supportive and empathetic response to a user who is struggling with feelings of despair and suicidal thoughts. The author emphasizes that these feelings are temporary and can be managed with time, patience, and a shift in perspective. They encourage the user to focus on their own psychological profile, triggers, and emotional well-being, and to seek help if needed. The author also emphasizes the importance of not identifying with negative thoughts and feelings, and of cultivating positivity and love in one's life. They suggest that by doing so, the user can overcome their current difficulties and emerge stronger, smarter, and more emotionally aware. The author also shares their own experiences with bipolar disorder and the importance of understanding one's own psychological profile in managing mental health issues.

User 3:

The author shares their personal experience of having suicidal thoughts and attempts, but ultimately deciding to live after realizing they wanted to continue living. They emphasize the importance of making significant changes in one's life, such as starting sports, making friends, and forming relationships, as these changes helped them find a reason to carry on. The author encourages others to find their own reasons for living and to not give up hope, as suicide is not a solution. They suggest talking to friends or family when feeling down and remind others that there are people who care about them and that better times can come. The author also shares their own struggles with pain and illness, but emphasizes that they have overcome these challenges and that others can too. They encourage resilience and self-belief, reminding others that they are more than their pain and that they have already overcome a lot.

User 4:

The author, who identifies as a female, shares her personal struggles with thoughts of self-harm and grief, having lost a loved one. She expresses empathy towards the reader, offering support and encouragement, and emphasizes the importance of hobbies and self-improvement in coping with difficult times. She also warns the reader against suicide, stating that it could potentially cause irreparable harm to loved ones who are left behind, as they may have to live with the grief for the rest of their lives. The author concludes by acknowledging the reader's bravery for persevering through their struggles.

User 5:

The author on Reddit is providing empathetic and supportive responses to a user who is experiencing emotional distress, self-harm, and suicidal thoughts. The author expresses concern for the user's safety and encourages them to seek help from a therapist or care coordinator, even if they have had negative experiences with them in the past. The author also offers to listen and provide support to the user, acknowledges their feelings of loneliness and frustration, and suggests distractions like games to help cope with difficult emotions. The author also shows concern for the user's cat and asks if the user has anyone in their life they can reach out to for support. The author's language is understanding, non-judgmental, and supportive, and they express a desire to help the user through their difficult time.

Appendix C

Response generation examples

Zero-prompted user-2 for Mistral-7b-instruct-v0.3:

Dear Author,

I want to express my deepest gratitude for sharing your story and offering such heartfelt advice to those who may be struggling with similar feelings. Your resilience and empathy are truly inspiring.

In times of distress, it's essential to remember that self-care is not selfish, but necessary. I encourage you to continue advocating for self-improvement and self-love. Activities like spending time outdoors, connecting with new people, pursuing hobbies, joining a gym, or reconnecting with old friends can help uplift your spirits.

Remember, it's okay to be a little selfish and treat yourself to something you enjoy, like a meal at your favorite restaurant. Changing one's mindset can be challenging, but remember, there is always a way out, and committing suicide is not the only decision.

I hope you continue to find strength in your journey and that your words of encouragement reach those who need them. If you or someone else is struggling, please remember that help is available. In the UK, you can reach out to the Emergency Services at 999, the Samaritans at 116 123, or text SHOUT at 85258.

Take care and stay strong. I believe things will get better for you, but only if you choose to keep living.

Warm regards

Chain-of-thought prompted user-2 for Mistral-7b-instruct-v0.3:

Dear Author,

I want to commend you for your courage in sharing your story and offering empathy and advice to those who may be going through a similar situation. Your words are a testament to your strength and resilience, and they can truly make a difference in someone's life.

I understand that you've been through a lot, especially with the feelings of abandonment and worthlessness caused by your father's abusive behavior. It's important to remember that these feelings are not your fault, and it's okay to seek help and prioritize your own well-being.

You've offered some great suggestions for self-care and self-improvement, such as going outside, talking to new people, taking up hobbies, joining a gym, or reconnecting with old friends. These are all wonderful ideas, and I encourage you to continue to find activities that bring you joy and help you feel more connected.

You've also emphasized the importance of being a little selfish and doing things for oneself, such as treating oneself to a favorite restaurant. This is a crucial reminder that taking care of oneself is not selfish, but necessary for overall well-being.

I know that changing one's mindset can be difficult, but I want to remind you that there is always a way out, and committing suicide is not the only decision. It's important to remember that there are people who care about you and want to help.

I want to express my hope that things will get better for you, but only if you choose to keep living. If you ever feel overwhelmed or need someone to talk to, please don't hesitate to reach out to the Samaritans at 116 123, or text SHOUT at 85258. If you feel that you are in immediate danger, please call the Emergency Services at 999.

Take care of yourself, and know that you are not alone. You are strong, and you have the power to make a positive change in your life.

Warm regards

Zero-prompted user-2 for Tulu-2-DPO-7b:

Dear Author,

I am sorry to hear that you have been struggling with feelings of abandonment and worthlessness, which have led you to contemplate suicide. I want you to know that you are not alone, and there are people who care about you and want to help.

As someone who has recently been in a similar situation, I understand how difficult it can be to cope with these feelings. However, I want to encourage you to focus on self-care and self-improvement. This can include activities such as going outside, talking to new people, taking up hobbies, joining a gym, or reconnecting with old friends. It is important to be a little selfish and do things for yourself, such as treating yourself to a favorite restaurant.

Remember, changing your mindset can be difficult, but it is always possible. I want to remind you that there is always a way out, and committing suicide is not the only decision. I hope that things will get better for you, but only if you choose to keep living.

If you are in need of immediate support, please do not hesitate to reach out to the following helplines: Emergency Services: 999, Samaritans helpline: 116 123, or SHOUT texting helpline: 85258. These services are available 24/7 and are free and confidential.

Take care of yourself, and know that you are not alone.

Sincerely

Chain-of-thought prompted user-2 for Tulu-2-DPO-7b:

Dear author,

I can understand how difficult it must be for you to cope with feelings of abandonment and worthlessness caused by your father's abusive behavior. It takes a lot of courage to share your story and offer empathy and advice to others who may be going through a similar situation.

I want you to know that you are not alone, and there is always hope for a better future. It's great that you are focusing on self-care and self-improvement by suggesting activities like going outside, talking to new people, taking up hobbies, joining a gym, or reconnecting with old friends. It's important to be a little selfish and do things for yourself, such as treating yourself to a favorite restaurant.

Remember, changing your mindset can be difficult, but it's not impossible. You have the strength and resilience to overcome these challenges. I want to encourage you to keep reaching out for help and support, whether it's through self-help resources or professional help.

If you ever feel like you are at risk of harming yourself, please don't hesitate to call the emergency services on 999, or the Samaritans on 116 123. They are there to support you 24/7.

I hope that things will get better for you, and I believe that you have the power to choose to keep living. You are strong, and you deserve to live a fulfilling life.

Sincerely

Appendix D

Results

Full results are available here: <https://github.com/aiden-os/Dissertation>

Appendix E

Code

Full code is available here: <https://github.com/aiden-os/Dissertation>

Listing E.1: The function for allocating a user's risk level based on their highlights, and ratio of supportive sentences

```
def calculate_user_risk_rating(user_sentences, supportiveness_ratio):
    if user_sentences.empty:
        return 0

    rating_counts = user_sentences['rating'].value_counts()
    max_rating = user_sentences['rating'].max()
    print(user_sentences)
    if supportiveness_ratio >= 0.9:
        if max_rating < 4:
            return 1

    ratings_of_4_or_5 =
        user_sentences[user_sentences['rating'].isin([4, 5])]
    print(ratings_of_4_or_5)
    all_other_ratings_are_2 =
        user_sentences[user_sentences['rating'] != 2].shape[0] ==
        len(ratings_of_4_or_5)
    if supportiveness_ratio >= 0.87 and len(ratings_of_4_or_5) == 1
        and all_other_ratings_are_2:
        return 1
    if supportiveness_ratio >= 0.87 and len(user_sentences) in {1,2}:
        return 1

    if 4 in user_sentences['rating'].values or 5 in
        user_sentences['rating'].values:
        print("hi")
        if len(ratings_of_4_or_5) >= 1:
            print("hi2")
            if user_sentences['rating'].mean() > 4.5:
                return 5
            if user_sentences['rating'].mean() > 3.5:
                return 4
```

```
        else:
            return max(3, int(user_sentences['rating'].mean()))

num Rated_3 = (user_sentences['rating'] == 3).sum()
num_sentences = len(user_sentences)
if num Rated_3 > (num_sentences * 0.3):
    return 3
return int(user_sentences['rating'].mean())
```